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$$z^2 - (1 - i)(z - 2) = 0$$

$$\Leftrightarrow z^2 + (-1 + i)z + (2 - 2i) = 0$$

$$\begin{aligned} D &= (-1 + i)^2 - 4 \cdot 1 \cdot (2 - 2i) \\ &= i^2 - 7 + 6i \\ &= -8 + 6i \end{aligned}$$

$$\begin{cases} x^2 - y^2 &= -8 \\ 2xy &= 6 \end{cases} \Leftrightarrow \begin{cases} x^2 + (-y^2) &= -8 \\ x(-y) &= -9 \end{cases}$$

$$\lambda^2 + 8\lambda - 9 = 0$$

$$\Delta = 8^2 - 4 \cdot 1 \cdot (-9) = 10^2$$

$$\lambda_{1,2} = \frac{-8 \pm 10}{2}$$

$$\lambda_1 = x^2 = 1$$

$$\lambda_2 = -y^2 = -9$$

$$x + yi \in \{1 + 3i, -1 - 3i\}$$

$$z = \frac{-(-1 + i) \pm (1 + 3i)}{2}$$

$$z_1 = \frac{2 + 2i}{2} = 1 + i$$

$$z_2 = \frac{-4i}{2} = -2i$$

References